


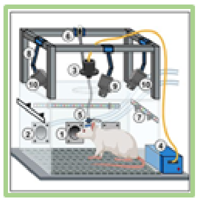
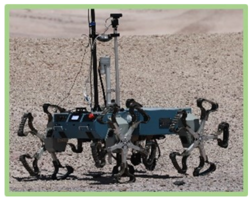

Trevor R. Smith

tresmi42@gmail.com   814-691-4529
Google Scholar: https://scholar.google.com/citations?user=cfqfknwAAAAJ&hl=en
Personal Website: https://trevorsmith42.github.io/
LinkedIn: https://www.linkedin.com/in/trevor-smith-05173b147/

Research Profile

I research the scalable co-emergence of collective intelligence, morphology, and behavior to form a single coherent machine. I have developed self-organizing modular swarm systems—from multicellular and multi-arm robots to intuitive and adaptive human–machine interfaces for heavy equipment. Across my Ph.D. and current MIT postdoctoral research, I have designed 20+ platforms comprising 200+ physical robots, published 7 first-author papers, and received 4 presentation awards.

Research Interests: Swarm and modular robotics; self-organization; collective and morphological intelligence; decentralized control; adaptive morphology

Applications: Field, construction, agricultural, and space robotics

Education

Aug. 2026 — **Ph.D., Mechanical Engineering**, West Virginia University (WVU), Morgantown, WV
National Science Foundation Graduate Research Fellow
Dissertation: *Swarm of One: Self-Organization in Multicellular Robots*
Advisor: Dr. Yu Gu
Committee: Drs. Y. Gu, N. Szczecinski, S. Yakovenko, A. Halasz, A. Mishra

Dec. 2021 — **B.S., Mechanical Engineering**, WVU, Morgantown, WV
B.S., Aerospace Engineering
Minor: Computer Science | Summa Cum Laude (4.0 GPA)

Fellowships

2025–2026 — **NASA West Virginia Space Grant Consortium Graduate Research Fellowship**
Competitive one-year state fellowship supporting NASA mission-aligned robotics research.

2022–2025 — **National Science Foundation Graduate Research Fellowship**
Prestigious three-year national fellowship supporting independent doctoral research in STEM disciplines.

Grants Contributed To

2024–2026 — **NSF FRR EAGER: Towards Bottom-Up Problem-Solving with Loopy, A Multicellular Robot**
Funding: $234,544 | Award No. 2422270
Assisted in drafting proposal sections on *Technical Approach*, *Preliminary Work*, *Intellectual Merit*, and *Broader Impacts*.

2021–2026 — **USDA/NIFA NRI: Stickbug — An Effective Co-Robot for Precision Pollination**
Funding: $750,361 | Award No. 2022-67021-36124
Supported proposal preparation by contributing text and figures for the *Technical Approach* and *Preliminary Work* sections.

Awards and Achievements

Apr. 2026 — **Outstanding Graduate Student Research Award**
Recognized for demonstrating exemplary academic achievement and research excellence.

WVU Rapid Fire Presentation, 1st Place
Recognized for effectively communicating Ph.D. research to a general engineering audience.

Apr. 2025 — **WVU 3-Minute Thesis Competition, 2nd Place**
Recognized for effectively communicating Ph.D. research to a general audience.

July 2024 — **Living Machines Conference, Best Presentation Award**
International biomimetics and biohybrid robotics conference; awarded for excellence in research communication.

Dec. 2021 — **WVU Undergraduate Research Symposium Winner — Science & Technology Category**

2017–2021 — **WVU Undergraduate Honors and Research Scholarships**
Includes Statler Research, Robotics, and Everette C. Dubbe Scholarships; WVU Honors College graduate.

Academic Appointments and Publications

Cross-Collaborations: † University of Texas at Dallas Department of Bioengineering, ‡ West Virginia University Department of Neuroscience, § University of Florida Department of Industrial and Systems Engineering

Aug. 2026–Present — **Postdoctoral Associate**, MIT, Cambridge, MA
Developing intuitive human–robot interfaces for controlling high-DoF heavy machinery, combining haptic feedback, human motion and intent, and robotic control. Current work investigates naturalistic control of excavators with tiltrotators as a platform for human-centered teleoperation and shared autonomy.

Aug. 2022–2026 — **Graduate Research Assistant**, WVU, Morgantown, WV
Research in swarm and bio-inspired robotics focused on self-organization, mechanical intelligence, and adaptive morphology. Developed the *Loopy* multicellular robot to study morphogenesis-driven collective behavior. Extended this work to the six-armed Stickbug co-robot for precision pollination, demonstrating scalable mechanical intelligence. Applied work to distributed automation systems for rodent behavioral neuroscience, integrating real-time sensing, actuation, and feedback.

May 2019–2022 — **Undergraduate Research Assistant**, WVU, Morgantown, WV
Investigated multi-agent robotics and developed an experimental testbed for tabletop swarms integrating motion capture, video projection, and low-friction environments. Participated in an NSF Research Experience for Undergraduates (REU) on flocking-based localization using adaptive Boids rules for cooperative positioning.

Manuscripts Under Review

2026 — Bamani Beeri, E., Buzzatto, J., **Smith, T.R.**, Baek, J., Laban, G., and Krebs, H.I., *Human gesture recognition beyond clip accuracy: A critical perspective on multimodal AI from sensing to action*. Artificial Intelligence Review.

Bamani, E.B., Jeong, S., **Smith, T.R.**, and Krebs, H.I., *Comparing soil dynamic models for the estimation of reaction forces on a virtual excavator bucket*. Submitted to the 5th Workshop on the Future of Construction: Collaborative Robots for Fabrication, Manufacturing, and Inspection, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026), Pittsburgh, PA, USA.

Peer-Reviewed Journal Articles

2025

- [†‡] **Smith, T.R.**, Smith, T.J., Szczecinski, N.S., Yakovenko, S., Gu, Y., *Cellular plasticity model for self-organized phenotypes in multi-cellular robots*. npj Robotics
  `https://doi.org/10.1038/s44182-025-00039-y`

2024

- [†] Smith, T. J., **Smith, T. R.**, Faruk, F., Bendea, M., Kumara, S. T., Capadona, J. R., Hernandez-Reynoso, A. & Pancrazio, J. J., *Real-Time Assessment of Rodent Engagement Using ArUco Markers: A Scalable and Accessible Approach for Scoring Behavior in a Nose-Poking Go/No-Go Task*. Eneuro
  `https://doi.org/10.1523/ENEURO.0500-23.2024`
- [†] Smith, T. J., **Smith, T. R.**, Faruk, F., Bendea, M., Kumara, S. T., Capadona, J. R., Hernandez-Reynoso, A.& Pancrazio, J. J., *A Real-Time Approach for Assessing Rodent Engagement in a Nose-Poking Go/No-Go Behavioral Task Using ArUco Markers*. Bio-protocol
  `https://doi.org/10.21769/BioProtoc.5098`

2023

- [§] **Smith, T.**, Chen, Y., Hewitt, N., Hu, B., Gu, Y., *Socially Aware Robot Obstacle Avoidance Considering Human Intention and Preferences*. International Journal of Social Robotics
  `https://doi.org/10.1007/s12369-021-00795-5`

Refereed Conference Proceedings

2026

- Rijal, M., Shrestha, R., **Smith, T.**, & Gu, Y. *Modeling Branches for Active Manipulation using Iterative Parameter Estimation*. International Conference on Intelligent Robots and Systems (IROS) **Accepted**

2025

- Shrestha, R., Rijal, M., **Smith, T.**, & Gu, Y. *FloPE: Flower Pose Estimation for Precision Pollination*. International Conference on Intelligent Robots and Systems (IROS)
  `https://doi.org/10.1109/IROS60139.2025.11247634`
- Rijal, M., Shrestha, R., **Smith, T.**, & Gu, Y. *Force-Aware Branch Manipulation To Assist Agricultural Tasks*. International Conference on Intelligent Robots and Systems (IROS)
  `https://doi.org/10.1109/IROS60139.2025.11246611`
- [†] Smith, T. J., Tahmasebi, G., **Smith, T. R.**, Solis, E., Haghighi, P., Pancrazio, J.J., *Modular Tracking System for Treadmill-Based Rodent Experiments*. 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)
  `https://doi.org/10.1109/EMBC58623.2025.11254127`
- **Smith, T.** and Gu, Y. *Loopy Movements: Emergence of Rotation in a Multicellular Robot*, 2025 IEEE International Conference on Robotics and Automation (ICRA)
  `https://doi.org/10.1109/ICRA55743.2025.11128053`

2024

- **Smith, T.**, Rijal, M., Tatsch, C., Butts, R. M., Beard, J., Cook, R. T, Chu, A., Gross, J., Gu, Y., *Design of Stickbug: a Six-Armed Precision Pollination Robot*, 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
  `https://doi.org/10.1109/IROS58592.2024.10801406`
- [†‡] **Smith, T.R.**, Smith, T.J., Szczecinski, N.S., Yakovenko, S., Gu, Y. *Cellular Plasticity Model for Bottom-Up Robotic Design*. In Biomimetic and Biohybrid Systems, Living Machines
  `https://doi.org/10.1007/978-3-031-72597-5_18`

2023 **Smith, T.**, Butts, R. M., Adkins, N., & Gu, Y. *Swarm of One: Bottom-Up Emergence of Stable Robot Bodies from Identical Cells*. In 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
`https://doi.org/10.1109/IROS55552.2023.10342118`

2022 **Smith, T.**, Gutierrez, E., Bredu, J., Gu, Y., Gross, J., *Cooperative and Coordinated Localization of Swarm Robots using Adaptive Boids Rules*, Proceedings of the 35th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+)
`https://doi.org/10.33012/2022.18560`

2021 § Chen, Y., **Smith, T.**, Hewitt, N., Gu, Y., & Hu, B. *Effects of Human Personal Space on the Robot Obstacle Avoidance Behavior: A Human-in-the-loop Assessment*. Proceedings of the Human Factors and Ergonomics Society Annual Meeting
`https://doi.org/10.1177/1071181321651098`

Datasets

2024 Rijal, M., **Smith, T.**, Tatsch, C., Beard, J., Chu, A., Butts, M., Waterland, N., Gross, J., Gu, Y., *Bramble Flower Detection and Classification Dataset for Precision Pollination*, IEEE Dataport
`https://dx.doi.org/10.21227/b58f-jt53`

Presentations and Demonstrations

2025 **NSF–FRR/NRI Annual Meeting**, Alexandria, VA, USA.
Demonstrated *Loopy: A Multicellular Robot for Bottom-Up Problem Solving*.

IEEE International Conference on Robotics and Automation (ICRA), Atlanta, GA, USA.
Presented and demonstrated *Loopy Movements: Emergence of Rotation in a Multicellular Robot*.

WVU 3-Minute Thesis Competition. *2nd Place Winner*.

2024 **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, Abu Dhabi, UAE.
Presented *Stickbug: A Six-Armed Precision Pollination Robot*.

Living Machines Conference, Chicago, IL, USA.
Presented *Cellular Plasticity Model for Bottom-Up Robotic Design*. **(Best Presentation Award)**

NSF–FRR/NRI Annual Meeting, Baltimore, MD, USA.
Demonstrated *Stickbug: Co-Robot for Precision Pollination*.

2023 **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, Detroit, MI, USA.
Presented and demonstrated *Swarm of One: Bottom-Up Emergence of Stable Robot Bodies from Identical Cells*.

2022 **ION GNSS+**, Denver, CO, USA.
Presented *Cooperative and Coordinated Localization of Swarm Robots Using Adaptive Boids Rules*.

2021 **WVU Fall Undergraduate Research Symposium**, Morgantown, WV, USA.
Presented *Gemini Telepresence Robot System Design*. **(Best Presentation Award)**

Media Coverage

Research featured in national and international outlets.

- Feb. 2026 — **CBS: Visioneers With Zay Harding** — Season 2 Episode 13: *"Machines With Purpose"*
- Jul. 2024 — **ASME** — *"Six-Armed Robot Stickbug Works as a Precision Pollinator."*
- Feb. 2024 — **The Conversation** — *"Swarm of One Robot Is a Single Machine Made Up of Independent Modules."*
- Jul. 2023 — **Scientific American** — *"Robotic Bees Could Support Vertical Farms Today and Astronauts Tomorrow."*
- 2021–2023 — **IEEE Spectrum Robotics Blog, Fast Company, and WVU Today** — Coverage of Loopy and Stickbug projects in multiple issues of IEEE's "Video Friday" series and WVU research highlights.

Teaching

Hands-on experience, freedom to explore, and the freedom to make mistakes are critical to learning and innovation in robotics. By equipping students with technical expertise and problem-solving skills, I inspire future roboticists and prepare them for success.

Courses

- 2025 — **WVU, MAE 412: Mobile Robotics - Design of Education Robot Platform**
  Designed, fabricated, and programmed 20 state-of-the-art mobile and manipulator robots for educating students on navigation, sensor fusion, and kinematics.
- Summer 2024 — **WVU, MAE 460: Automatic Controls - Guest Lecturer**

Student Mentorship

- 2026–Present — **MIT Undergraduate Research Mentorship (2 students)**
  Mentored Austin Reinhardt and Kendall Binkley in robotics research, experimental analysis, and scientific writing, resulting in two papers submitted to the IEEE MIT Undergraduate Research Technology Conference.
- 2023–2026 — **WVU Graduate and Undergraduate Research Mentorship (2 master's and 11 undergraduate students)**
  Provided research and technical mentorship across mechanical engineering, computer science, aerospace engineering, electrical engineering, and robotics. Graduate mentees included Robert Tyler Cook (M.S. in Mechanical Engineering, 2026) and Jaleen Beeman (M.S. student in Computer Science). Undergraduate mentees achieved peer-reviewed co-authorship, leadership in the University Rover Challenge, graduate admissions, and positions at Starlink, Blue Origin, and MITRE. Selected mentees include Logan Gold, a Ruby Fellow and URC team lead, and Camndon Reed, a Goldwater Scholar.
- 2022 — **NSF Research Experience for Undergraduates — Graduate Mentor**
  Trained 10+ students during a 10-week intensive program through guided projects in swarm robotics, experimental design, and scientific communication.

2019–2026 **University Rover Challenge Team Mountaineers — Graduate Mentor and Former CTO**
Mentored a 40+ member interdisciplinary team in system design, electronics, and software integration. The team placed 1st (2023) and 2nd (2024 and 2025) among 100+ international teams.

WVU Robotics Club — Graduate Mentor and Former President
Expanded the club from 6 to 30 active members and established a dedicated laboratory. Mentored students in mechanical design, manufacturing, and programming; the team qualified for the VEX-U World Championship annually from 2021–2025.

Professional Service

Memberships

2021–Present **Institute of Electrical and Electronics Engineers (IEEE)**

Reviewer

2022–Present **Reviewed 25+ submissions for leading robotics venues including:**
- *IEEE Robotics and Automation Letters* (11)
- *International Journal of Social Robotics* (6)
- *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (9)
- *IEEE International Conference on Robotics and Automation (ICRA)* (3)

Outreach

2019–2025 **VEX Robotics Competition (VRC) — Volunteer**
Roles: Scorekeeper and Judge across multiple high school and middle school events per year. Supported *50+ students per event* through field operations, scoring, match coordination, and informal STEM instruction that encouraged teamwork, design thinking, and interest in robotics.

K-12 Lab Visitations and Demonstrations
Inspired and introduced K-12 students to the field of robotics with hands-on robotics demonstrations to an average group size of 10-20 students, 5 times a year.